A RING WHOSE COMPLETED POLYNOMIAL RING IS NOT FLAT OVER R

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: This is the plain lemma; the separately delivered '-bis' card (L48290) counts a different author unit whose discussion begins at line 1042; the two statement intervals are disjoint.

Editorial note. The label lemma-completion-polynomial-ring-not-flat names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 1018-1030: for R countable, R[[x]] is isomorphic to R^N as an R-module; by Lemma countable-coherent (same chapter, delivered as card L48289) the ring map R -> R[[x]] is flat if and only if R is coherent
- The context exhibits a noncoherent countable ring: R = k[y, z, a_1, b_1, a_2, b_2, ...]/(a_i y + b_i z for all i) over a countable field k, not coherent because the ideal (y, z) is not a finitely presented R-module
- R[[x]] is the completion of R[x] by the principal ideal (x) (context closing sentence)

Source reference. Original reference 0ALB: sources/stacks-examples/pinned-original-source.tex, source lines 1001–1001. Prerequisite pointer invoked at 1043; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Let R be a countable ring. Observe that $R[[x]]$ is isomorphic to $R^{\mathbf{N}}$ as an R-module. By Lemma 0ALB we see that $R \to R[[x]]$ is flat if and only if R is coherent. There are plenty of noncoherent countable rings, for example

$$R = k[y, z, a_1, b_1, a_2, b_2, a_3, b_3, \ldots]/(a_1 y + b_1 z, a_2 y + b_2 z, a_3 y + b_3 z, \ldots)$$

where k is a countable field. This ring is not coherent because the ideal (y, z) of R is not a finitely presented R-module. Note that $R[[x]]$ is the completion of $R[x]$ by the principal ideal (x).

Lemma 0.1. *There exists a ring such that the completion $R[[x]]$ of $R[x]$ at (x) is not flat over R and a fortiori not flat over $R[x]$.*

Proof. See discussion above. □

Date: stacks-project:examples.tex:local-snapshot-20261012; GNU FDL 1.2 or later.